Lattice Trading Rules and Probability of Profit

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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Specify a trade rule.** Translate entry price, holding period, benchmark, target, and transaction costs into an unambiguous terminal profit event.
- **Derive lattice probabilities.** Convert a target-return inequality into a minimum up-move count and evaluate its real-world binomial tail probability.
- **Generalize the state space.** Represent an m-branch recombining lattice with multinomial state counts and explain its computational tradeoffs.

1 A Forecast Is Not Yet a Trading Rule

A price model describes possible states and their probabilities. A trading rule adds decisions: when to enter, how many shares to hold, what constitutes success, when to exit, and how costs are handled. Without those choices, "probability of profit" is not a well-defined quantity.

Suppose $n_0 > 0$ shares are purchased at the time-0 ask price $S_0^{\text{ask}} > 0$ dollars per share and liquidated after N steps at the bid price $S_N^{\text{bid}} > 0$. Let $K \geq 0$ dollars collect commissions and other modeled cash costs. With a continuously compounded annual benchmark rate r_b and horizon $T = N\Delta t$ years, one present-value profit measure is

$$\Pi_0 := e^{-r_b T} n_0 S_N^{\text{bid}} - n_0 S_0^{\text{ask}} - K. \tag{1}$$

Equation Eq. 1 compares all cash flows at time 0. An equivalent future-value comparison grows the entry cost and costs to time T. Mixing a discounted sale value with an undiscounted target would create inconsistent units.

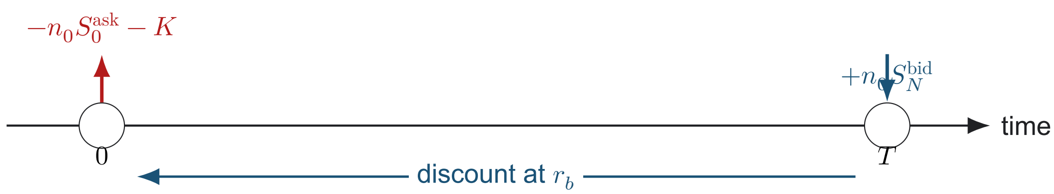

Figure 1: A terminal-exit long-equity rule represented as dated cash flows. Real execution introduces a bid–ask spread and costs; the benchmark rate converts the liquidation receipt to the entry date.

The benchmark encodes an opportunity-cost hurdle. Setting $r_b = 0$ asks whether nominal proceeds exceed cost. A Treasury-based or strategy-specific r_b asks whether the trade beats that alternative. Neither choice converts POP into investment advice; it only makes the comparison explicit.

2 Closed-Form Binomial Tail Rule

Use the real-world binomial lattice from L3b. Let $u > d > 0$ be the constant price factors, let $p \in [0,1]$ be the real-world up probability, and let the random up-move count be $K_N \sim \text{Binomial}(N,p)$. Ignoring bid–ask spread and fixed costs temporarily, the terminal node price is $S_0 u^{K_N} d^{N-K_N}$.

Define the present-value fractional return on initial share cost by

$$\rho_N(K_N) := e^{-r_b T}\frac{S_N}{S_0} - 1 = e^{-r_b T} u^{K_N} d^{N-K_N} - 1. \tag{2}$$

For a target $\rho_\star > -1$, the strict success event $\rho_N > \rho_\star$ becomes a threshold on the integer K_N (Thm. 2.1).

Theorem 2.1. Target-return probability in a binomial lattice

Let $u > d > 0$, $p \in [0,1]$, $N \in \mathbb{N}$, $\Delta t > 0$ years, $T := N\Delta t$, and $K_N \sim \text{Binomial}(N,p)$ under the real-world measure. Let the continuously compounded benchmark rate be $r_b \in \mathbb{R}$ per year and let $\rho_\star > -1$ be the dimensionless strict target for the present-value return in Eq. 2. Define

$$\tau(\rho_\star) := \frac{\log(1+\rho_\star) + r_b T - N\log d}{\log(u/d)}, \qquad k_{\min} := \lfloor \tau(\rho_\star) \rfloor + 1. \tag{3}$$

Then the real-world probability of exceeding the target is

$$\mathbb{P}(\rho_N > \rho_\star) = \sum_{k=\max(0,k_{\min})}^{N} \binom{N}{k} p^k (1-p)^{N-k}, \tag{4}$$

with value zero when $k_{\min} > N$ and value one when $k_{\min} \leq 0$.

Derivation. Start with $e^{-r_b T} u^k d^{N-k} - 1 > \rho_\star$. Move positive terms and take logarithms:

$$\begin{aligned} u^k d^{N-k} &> (1+\rho_\star)e^{r_b T}, \\ k\log(u/d) &> \log(1+\rho_\star) + r_b T - N\log d. \end{aligned}$$

Because $u/d > 1$, division preserves the inequality, so $k > \tau(\rho_\star)$. The smallest admissible integer is $\lfloor \tau \rfloor + 1$; summing the binomial masses above it gives Eq. 4. □

The strict inequality matters at a node lying exactly on the threshold. A non-strict goal $\rho_N \geq \rho_\star$ uses a ceiling rule instead. Numerical code should handle this boundary explicitly rather than rely on floating-point equality.

COMPANION NOTEBOOK

L4a: Cumulative Probability for a Lattice Exit Rule →

Estimate a real-world lattice, convert target returns into node thresholds, and compare the analytic binomial tail with enumerated states.

3 Terminal Rules and First-Passage Rules Differ

The theorem concerns a fixed terminal liquidation date. A take-profit or stop-loss instruction monitored during the holding period is path dependent: the first boundary crossing can occur before T, and two paths ending at the same terminal node may trigger different trades. Terminal node probabilities alone cannot evaluate such a rule.

Definition 3.1. First-exit time

Let S_j be the modeled share price at step $j \in \{0, \dots, N\}$, and let $L_j < U_j$ be lower and upper dollar boundaries. The first-exit time is the random index

$$\tau_{\text{exit}} := \min\{j \in \{1, \dots, N\} : S_j \leq L_j \text{ or } S_j \geq U_j\}, \tag{5}$$

with $\tau_{\text{exit}} := N$ if no earlier boundary is reached. A complete rule must also specify the liquidation price and action when a boundary is crossed or both instructions are affected by a price gap.

First-passage probabilities can be computed by propagating probability only through nodes that have not yet been absorbed. In actual markets, stop orders do not guarantee their trigger price, limit orders do not guarantee execution, and discrete monitoring can miss between-step crossings. Slippage and market impact make the execution model part of the strategy model.

4 From Two Branches to Many

A binary lattice compresses every step into two outcomes. An m-branch model uses positive factors $f_1, \dots, f_m$ with probabilities $p_1, \dots, p_m$, where $p_j \geq 0$ and $\sum_j p_j = 1$. After N independent steps, a state is indexed by the count vector $\mathbf{x} = (x_1, \dots, x_m)$ of nonnegative integers satisfying $\sum_j x_j = N$.

Proposition 4.1. Multinomial lattice state

Under the constant independent m-branch model, a count vector $\mathbf{x} \in \mathbb{Z}^m_{\geq 0}$ with $\sum_{j=1}^m x_j = N$ has terminal price and probability

$$S_N(\mathbf{x}) = S_0 \prod_{j=1}^{m} f_j^{x_j}, \qquad \mathbb{P}(\mathbf{X} = \mathbf{x}) = \frac{N!}{x_1! \cdots x_m!} \prod_{j=1}^{m} p_j^{x_j}. \tag{6}$$

The number of count vectors at level N is $\binom{N+m-1}{m-1}$. Distinct count vectors may coincide in price if the factors have additional multiplicative relationships.

More branches approximate a richer one-step distribution but do not automatically create heavy tails, volatility clustering, or accurate forecasts. Parameters, bin boundaries, and transition dependence still require out-of-sample validation. The computational notebook compares equal-mass and equal-width discretizations and checks that probability mass sums to one at each level.

COMPANION NOTEBOOK

L4a: N-Ary Recombining Lattices →

Discretize an empirical growth-rate distribution into multiple branches, build multinomial states, and visualize terminal prices and probabilities.

KEY TAKEAWAYS

In this lecture, we converted a lattice forecast into a complete terminal trade rule, derived its target-return probability, and separated terminal from first-passage exits.

- **Profit is a cash-flow event.** Entry, exit, benchmark, horizon, spread, fees, and the strictness of the target must be specified before POP exists.
- **Thresholds become count tails.** A monotone binomial terminal payoff converts a return hurdle into a minimum number of up moves and an exact real-world binomial tail.
- **Path dependence changes the state.** Stop-loss and take-profit rules depend on the first boundary crossing, so terminal node probabilities alone are insufficient.

Next, take the continuous-time limit and derive geometric Brownian motion, its lognormal price distribution, and its closed-form target probability.

ADDITIONAL READING

- Sheldon M. Ross, *Introduction to Probability Models*, 13th ed. (2023), for binomial, multinomial, and first-passage constructions.
- John C. Hull, *Options, Futures, and Other Derivatives*, 11th ed. (2021), for lattice state propagation and no-arbitrage applications.
- U.S. Securities and Exchange Commission, *Types of Orders*, for the practical distinctions among market, limit, and stop orders.